FACULTY OF ENGINEERING AND ARCHITECTURE
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Development and Optimization of Coil Designs for Kidney Cancer with Artificial Intelligence Assisted Pulsed Electromagnetic Field Systems

GRADUATION/DESIGN PROJECT

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Development and Optimization of Coil

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A GRADUATION/DESIGN PROJECT

by

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submitted to the Electrical and Electronics Engineering of

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ABSTRACT

Electromagnetic fields play an important role in biology and medicine as a stimulant and therapeutic. In recent years, the effects of pulsed electromagnetic fields on soft tissue injuries, skin ulcers, irregular bone fractures and degenerative nerve healing, cancer treatment and cancer spread have been studied. In this project, a customizable DMAT (Pulsed Electromagnetic Field) device to be used for cancer treatment will be developed. This device will target cancer cells specifically for different types of cancer and their regions and will offer the ability to select the appropriate waveform (sinusoidal, square and saw wave) to effectively destroy or reduce their functions. In addition, the most appropriate frequency value and electromagnetic field intensity to affect cancer cells will be determined and the appropriate duration for therapy will be defined. All these parameters can be managed via a Python-based artificial intelligence-supported interface that can be easily controlled by the user. Since the size, depth, temperature and magnetic field size of the surface where DMAT treatment is applied must be monitored, the physical data of the instant environment can be controlled throughout the treatment and can be easily observed by those who need treatment and those who apply the treatment thanks to the artificial intelligence and hardware components to be used. The DMAT device to be developed aims to reduce side effects and increase the rate of complete remission in patients by offering an alternative approach to existing treatment strategies in cancer treatment. flexibility of the device and its customization by taking into account different cancer types and patient characteristics will provide an important advantage that will increase treatment success. In this way, DMAT treatment can become a more effective and personalized treatment option for cancer patients.

Keywords:Pulsed Electromagnetic Field Therapy, Medical Device design, Coil, Microprocessor, DMAT system, Artificial intelligence

ÖZET

Elektromanyetik alanlar, biyoloji ve tıpta uyarıcı ve tedavi edici olarak önemli bir role sahiptir. Son yıllarda darbeli elektromanyetik alanların, yumuşak doku yaralanmaları, deri ülserleri, düzensiz kemik kırıkları ve dejeneratif sinir iyileşmesi, kanser tedavi ve kanser yayılımı üzerindeki etkileri çalışılmaktadır.Bu projede, kanser tedavisi için kullanılacak özelleştirilebilir bir DMAT (Darbeli Elektromanyetik Alan) cihazı geliştirilecektir. Bu cihaz, farklı kanser türlerine ve bulundukları bölgelere özgü olarak kanserli hücreleri hedef alacak ve etkili bir şekilde yok etmeye veya fonksiyonlarını azaltmaya yönelik olarak uygun dalga formunu (sinüzoidal, kare ve testere dalga) seçme yeteneği sunacaktır. Ayrıca, kanserli hücreleri etkilemek için en uygun frekans değeri ve elektromanyetik alan yoğunluğu belirlenecek ve terapi için uygun süre tanımlanacaktır. Tüm bu parametreler, kullanıcı tarafından kolayca kontrol edilebilecek bir Python tabanlı yapay zeka destekli arayüzü üzerinden yönetilebilecektir. DMAT tedavisi uygulanan yüzeyin büyüklüğü, derinliği, ortamın sıcaklığı ve manyetik alan büyüklüğünün takip edilmesi gerekli olduğundan, tedavi boyunca anlık ortamın fiziksel verileri kontrol edilebilecek ve kullanılacak yapay zeka ve donanımsal bileşenler sayesinde tedaviye ihtiyacı olan ve tedaviyi uygulayan tarafından kolaylıkla gözlemlenebilecektir.Geliştirilecek DMAT cihazı, kanser tedavisinde mevcut olan tedavi stratejilerine alternatif bir yaklaşım sunarak yan etkileri azaltmayı ve hastalarda tam remisyona ulaşma oranını artırmayı amaçlamaktadır. Cihazın esnekliği, farklı kanser türleri ve hasta özellikleri göz önüne alınarak özelleştirilebilmesi, tedavi başarısını artıracak önemli bir avantaj sağlayacaktır. Bu sayede, DMAT tedavisi kanser hastaları için daha etkili ve kişiye özgü bir tedavi seçeneği haline gelebilir.

Anahtar Kelimeler: Darberli Elektromanyetik Alan Terapi, Tıbbi Cihaz tasarımı,
Bobin, Mikroişlemci, DMAT sistemi, Yapay zeka

ACKNOWLEDGMENTS

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SYMBOLS

(°) **Temperature**

I. INTRODUCTION

Cancer remains one of the leading causes of mortality worldwide, posing a significant threat to public health. Kidney cancer, in particular, stands out due to its metastatic potential and resistance to conventional therapies, which often result in severe side effects and reduced patient quality of life. In this context, innovative non-invasive treatments are critical for improving both survival rates and post-therapy well-being.

Electromagnetic fields (EMFs) have demonstrated regulatory effects on cellular processes, including the control of cell growth, proliferation, apoptosis, and differentiation. Recent studies report that well-tuned Pulsed Electromagnetic Fields (PEMFs) can exert anti-proliferative and pro-apoptotic effects on various cancer cell lines, offering a promising modality for targeted therapy.However, the clinical efficacy of PEMF treatment critically depends on the geometry, frequency, intensity, and waveform of the applied field, all of which are dictated by the underlying coil design.

In this project, we develop a customizable PEMF (Pulsed Electromagnetic Field Therapy) device specifically aimed at kidney cancer treatment. The system integrates:

- A range of coil geometries (array coil structures with NiTi alloy wire) optimized via AI-supported algorithms to maximize magnetic flux density and field uniformity while operating within safe exposure limits (ICNIRP, ISO 10974).
- Selectable waveforms (sinusoidal, square, sawtooth) and frequencies (0–150 Hz) to identify the optimal parameters for inducing apoptosis in renal carcinoma cells.
- Real-time monitoring of treatment surface area, temperature, and field intensity through Hall-effect and temperature sensors, with data visualization and control via a Python-based GUI.
- Segmentation network development for processing microscopic images of treated cancer cells, enabling automated evaluation of treatment efficacy and facilitating a digital twin of the PEMF system.

1.1. Objectives and Scope

- **Objective 1:** Quantitatively analyze the effects of coil winding number, wire thickness, and coil diameter on magnetic field distribution and energy consumption using CST Studio Suite simulations and laboratory measurements.
- **Objective 2:** Fuse simulation and experimental datasets to identify the optimal coil parameter set for maximizing therapeutic impact on kidney cancer cells.
- **Objective 3:** Evaluate the AI-optimized coil designs in phantom and cell culture models of kidney cancer, assessing temperature elevation, field penetration, and cell viability changes over time.

By adopting a hybrid methodology that couples numerical simulations, AI-driven optimization, and experimental validation, this thesis seeks to establish a robust framework for personalized PEMF therapies in oncology.

1.2. Thesis Structure

This thesis is organized as follows:

Chapter 1 – Introduction

This chapter outlines the motivation behind the project, focusing on kidney cancer's resistance to conventional therapies and the growing interest in Pulsed Electromagnetic Field (PEMF) systems. It discusses the biological effects of EMFs on cellular processes and introduces the aim of developing a customizable, AI-assisted PEMF device (DMAT) targeting renal carcinoma cells.

Chapter 2 – Requirements and Design Methodology

This chapter presents the overall system design process, including data acquisition from sensors and simulations, preprocessing steps, and the implementation of AI-based optimization algorithms. The methodology integrates software, hardware, and clinical safety considerations to create a robust and scalable therapeutic device.

Chapter 3 – Materials and Methods

The third chapter covers the theoretical background of magnetic field generation using coils, including Biot–Savart calculations, coil driver circuit designs, and system architecture. It also details the components used (sensors, microcontrollers, power supplies), and the test plan employed to verify system performance and accuracy.

Chapter 4 – Simulation and Results

This chapter includes CST Studio simulations and experimental results from phantom and cell culture models. It compares the performance of different coil geometries and frequencies. Simulations show magnetic field distribution and electric loss in tumor vs. healthy tissue, while experiments validate thermal and biological impacts. The 150 Hz application was shown to be the most effective.

Chapter 5 – GUI and Embedded System Design

The fifth chapter describes the development of the Python-based graphical user interface (GUI) and the embedded system based on STM32. It includes the logic and code for sensor integration, PWM signal control, real-time monitoring, and AI-assisted decision-making mechanisms for adaptive field modulation.

Chapter 6 – Discussion in Terms of Constraints

This section discusses ethical, health, legal, and security constraints. Compliance with international standards such as ICNIRP (for electromagnetic exposure), HIPAA/GDPR (data privacy), and IEC 60601 (medical device safety) is addressed. Considerations such as electromagnetic interference protection and hardware isolation are emphasized.

Chapter 7 – Standards

All relevant international standards used in the project are reviewed. These include:

- **ISO 10974** – Electromagnetic safety limits
- **IEC 60601-1 / 60601-1-2** – Medical electrical device safety and EMC
- **IEEE 11073** – Interoperability in clinical systems
- **ISO 13485** – Medical device quality management
- **IEC 62304** – Software lifecycle in medical systems
- **ICNIRP 2020 –** EMF exposure guidelines
- **HIPAA / GDPR** – Data protection and privacy

Chapter 8 – Summary and Conclusion

Key findings are summarized: the DMAT system successfully integrates AI, real-time feedback, and simulation-backed design to offer a non-invasive, patient-specific therapy approach. Experiments confirm selective thermal and biological effects on cancer cells. The system shows promise for future multi-cancer use with adaptive control capabilities.

Chapter 9 – Appendix D – Standards

This appendix provides extended descriptions of the standards applied in the development and validation of the PEMF system, and how compliance was achieved during each phase.

II. REQUIREMENTS AND DESIGN METHODOLOGY

1. Data Collection

In this study, the data collection phase involved recording electromagnetic field (EMF) data obtained from kidney cancer cell experiments. In addition, real-time IoT sensor data—including temperature, magnetic field intensity, and tissue depth measurements—were systematically gathered to support the experimental setup. Clinical response data for various waveforms such as sinusoidal, square, and sawtooth, as well as different frequencies, were also recorded to evaluate the treatment effectiveness under diverse conditions.

2. Preprocessing

The acquired EMF data were standardized to address tissue heterogeneity, ensuring consistent analysis across different experimental conditions. To maintain the highest level of data privacy, patient information was anonymized in strict compliance with HIPAA and GDPR standards. Furthermore, noise filtering techniques were applied to the dataset to optimize it for the subsequent training of the AI model, resulting in a robust and reliable data pre-processing pipeline.

3. Model Design (Python-Based)

AI Optimization System:

During the model design phase, a Python-based AI optimization system was developed to simulate and optimize coil designs for effective treatment. This system is capable of recommending appropriate waveforms, frequencies, and magnetic field intensities based on specific tumor profiles. In addition, it employs a dynamic real-time feedback mechanism that automatically adjusts the therapy parameters to enhance treatment efficiency.

Integration:

The developed model integrates computational techniques such as the Finite Element Method (FEM) to accurately compute the EMF distribution in tissue. A clinician-friendly user interface was also incorporated, enabling healthcare professionals to customize parameters such as tumor depth and safety constraints. This seamless integration ensures that both the computational analysis and practical application are aligned for optimal therapeutic outcomes.

4. Evaluation Metrics

To assess the system's performance, its efficacy is evaluated by measuring tumor size reduction and assessing cancer cell viability. Safety evaluations are conducted using thermal imaging and biomarker analyses to monitor any potential impact on healthy tissue. Moreover, the accuracy of the AI system is verified by comparing the recommended parameters with actual clinical outcomes. The overall design also adheres to rigorous standards such as ISO 10974 for EMF safety and IEC 60601 for medical device compliance, ensuring that the system meets both clinical and regulatory requirements.

5. Performance & Scalability

Real-time sensor processing within the system allows for the adaptive adjustment of therapeutic parameters, thereby ensuring consistently high performance. The system has undergone scalability testing using modified datasets to validate its applicability to other cancer types, including liver and lung cancers. Furthermore, embedded system optimization through platforms such as Raspberry Pi and Arduino has been implemented, contributing to the overall efficiency and robustness of the therapy system.

6. User Interface (Python Dashboard)

A comprehensive Python-based dashboard was developed to visualize real-time EMF data and tumor metrics, facilitating immediate insights into the treatment process. This dashboard empowers users to customize therapy parameters, such as waveform selection and frequency adjustments, thereby enabling a more personalized treatment approach. Integrated safety alerts, which notify the user in cases of overheating or excessive EMF levels, further enhance patient security. Additionally, role-based access control has been implemented to differentiate functionality between clinicians and patients.

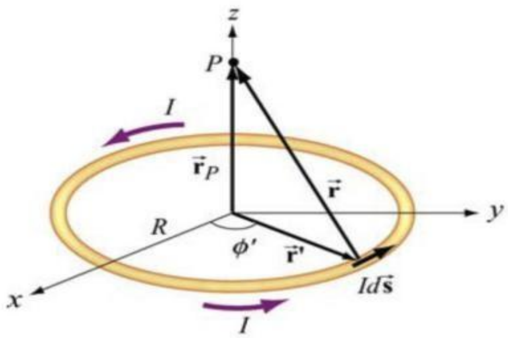
III. MATERIALS AND METHODS

3.1 Equations and Mathematical Expressions

In PEMF systems, the Biot-Savart Law is a fundamental tool for calculating the magnitude and direction of magnetic fields created by current-carrying wires or coils. This law ensures that the magnetic field is optimized in the design of devices, creating the right effects in targeted areas.

$$\vec{B} = \frac{\mu_0}{4\pi} \int_{wire} \frac{I\, d\vec{l} \times \hat{r}}{r^2}.$$

For a wire coil of radius R and N wire turns, the magnetic field along the vertical axis through the center of the coil is:



$$B = 2\frac{\mu_0}{2}\frac{N I a^2}{\left[a^2 + \left(\frac{a}{2}\right)^2\right]^{3/2}} = \mu_0 \frac{8NI}{5a\sqrt{5}}$$

Figure 1:Biot-Savart Equations

When we look at the formula, while the magnetic field size at the midpoint is directly proportional to the number of turns of the coil, it is inversely proportional to the distance to the midpoint.

3.2. Coils

Coils have an important place in the scope of the project. The reason for this is that electromagnetic fields used for therapeutic purposes are formed by coils. A coil is an electrical and electronic component in which an electric current-carrying wire is wound together in coils. It is often used to create a magnetic field or induce an electric current. A coil consists of two basic components, usually a core or coreless. The core is made of material used to concentrate or direct the magnetic field. Common core materials used in coils may include iron, ferrite or air. In coreless coils, the magnetic field is usually created by the surrounding medium. Coils or inductors are a two terminal passive electronic component. Coils are represented by the letter "L" for short. Inductors are made of wire that has the property of inductance, that is, the inductors oppose the flow of current. The unit of inductance is Henry (H).

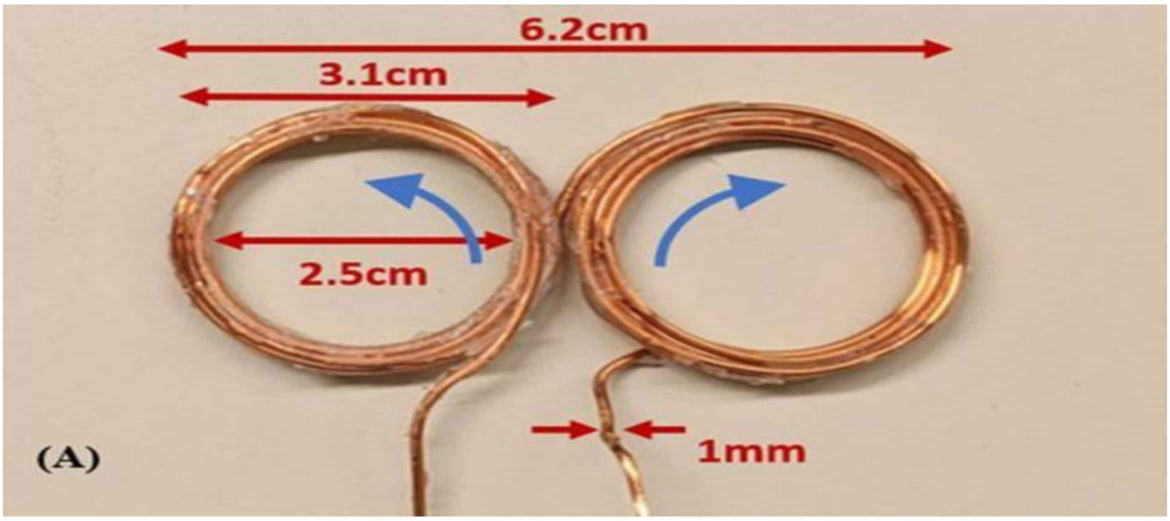


Figure2: Coil

The factors influencing the magnetic field of coils include:

Current Intensity: The strength of the magnetic field generated by a coil is directly proportional to the intensity of the electric current passing through it. Higher current intensity results in a stronger magnetic field.

Number of Turns: Increasing the number of turns in the coil increases the intensity of the magnetic field. More turns lead to a denser magnetic field and affect its distribution.

Core Material: Cores within coils influence the magnetic field. Different core materials, such as iron, ferrite, or air, interact differently with the magnetic field. For instance, an iron core enhances the magnetic field, while materials like air or ferrite have less impact on the magnetic field strength.

3.2.1 Coil driver circuit

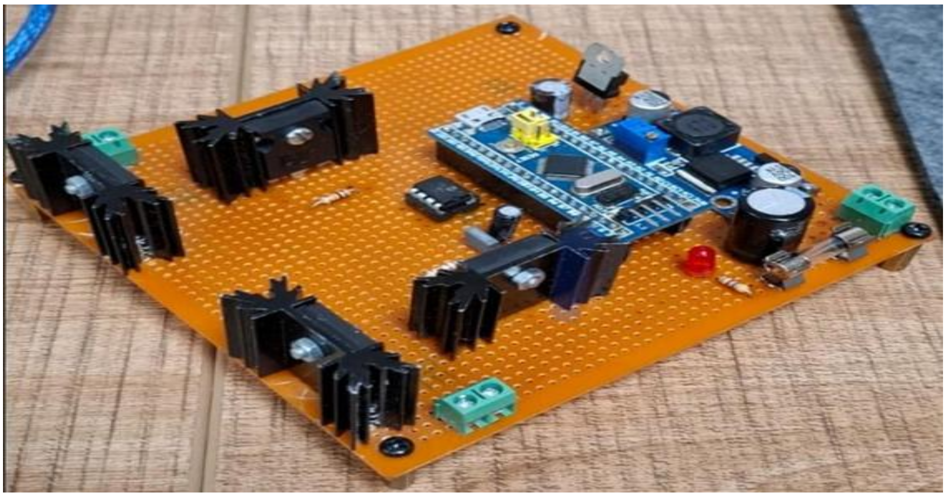


Figure2.1: Coil driver circuit

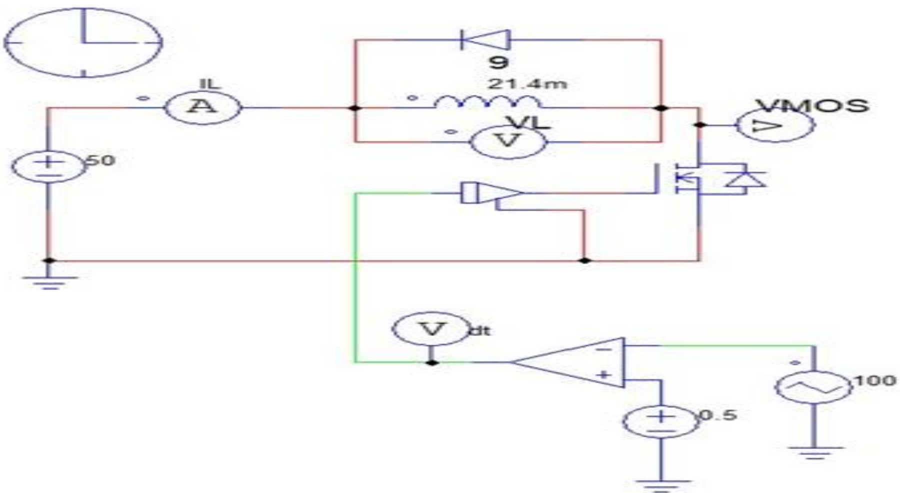


Figure2.2: Coil driver circuit schematic

The diagram shows the topology for driving an inductive load (coil) with a switching control element (MOSFET) and a freewheeling diode.The bottom section contains an op-amp-based driver circuit for error detection and a PWM generator, allowing precise control of the electric current applied to the coil via pulse-width modulation.In PEMF applications, this circuit ensures that the desired pulse frequency and amplitude are precisely delivered to the coil, allowing the therapeutic magnetic field to be optimized at the cellular level.

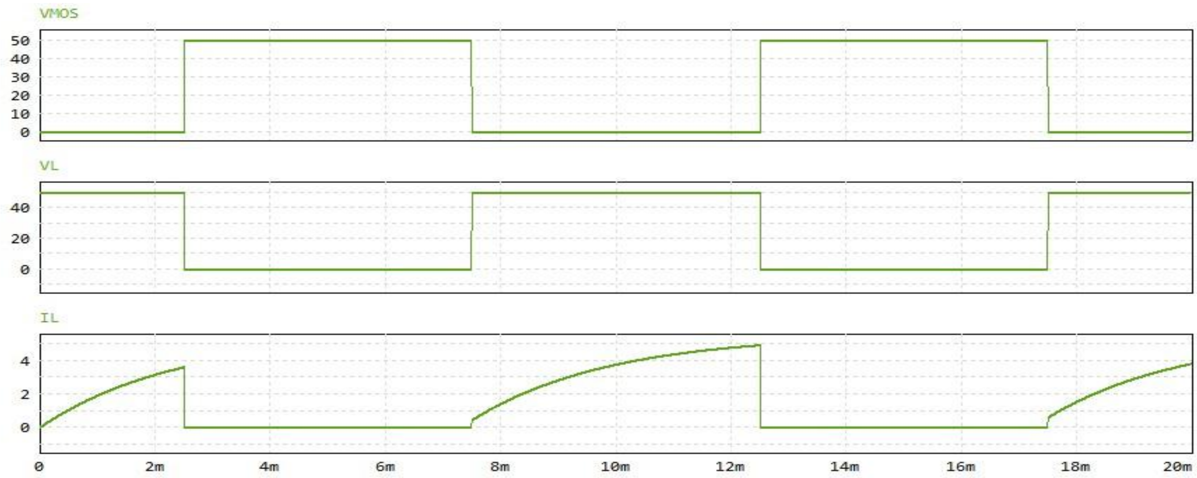


Figure2.3: VMOS, VL and IL Time Graphs

The upper graph shows that the input voltage (VMOS) applied to the MOSFET varies in regular square waves between 0 and 50 V depending on the PWM period.The middle graph (VL) reveals that the voltage across the coils is approximately 50 V when the MOSFET is turned on and decreases to 0 V when it is turned off, reflecting the energy storage and discharge cycle in the inductor.The lower graph (IL) shows that the coil current increases linearly with each on-off cycle and decreases to zero, enabling the inductive load to generate coherent magnetic field pulses at the target frequency in PEMF therapy.

3.3. System Components

3.3.1 AI Controlled System Components

a) Sensors

- **Magnetic Field Sensors:** Hall-effect sensors (KY-024) measure real-time magnetic flux density around each coil.
- **Temperature Sensors:** SHT31 temperature and humidity sensors prevent coil overheating by monitoring winding and phantom surface temperatures.
- **Current Sensors:** The ACS712 Current Sensor (-30 to +30 A) measures coil current values for real-time diagnostics.

- **Heart rate Sensors:** The AD8232 sensor monitors the patient's heart rhythm simultaneously.

b) Control Unit

- **Microcontroller:** The STM32F429ZI microcontroller manages sensor data acquisition, PWM generation control and safety interlocks.
- **AI Module Hardware:** A processor that can work with lightweight AI frameworks such as TensorFlow Lite, PyTorch Mobile (e.g. Nvidia Jetson, Raspberry Pi).

c) Artificial Intelligence Algorithms

- **Machine Learning:** Regression models (e.g., random forest, SVR) predict optimal current, frequency, and field uniformity metrics based on sensor inputs.
- **Deep Learning:** Multilayer neural networks (U-Net for segmentation, feedforward nets for dynamic load adaptation) optimize coil parameters under varying operational loads.

3.3.2 System Operating Principle

1. **Data Collection:** Instantaneous temperature, magnetic field intensity, current, and voltage data are captured via sensors and streamed to the microcontroller.
2. **Data Processing:** AI algorithms analyze incoming data to assess the current treatment state and calculate optimal coil parameters (current amplitude, pulse frequency, waveform shape).
3. **Control Signal Dispatch:** Computed parameters are transmitted to the coil driver via PWM signals from the microcontroller/FPGA, adjusting coil current, voltage, and frequency in real time.
4. **Feedback Loop:** Continuous monitoring enables closed-loop control; deviations from target field or temperature thresholds trigger AI-informed corrective actions.

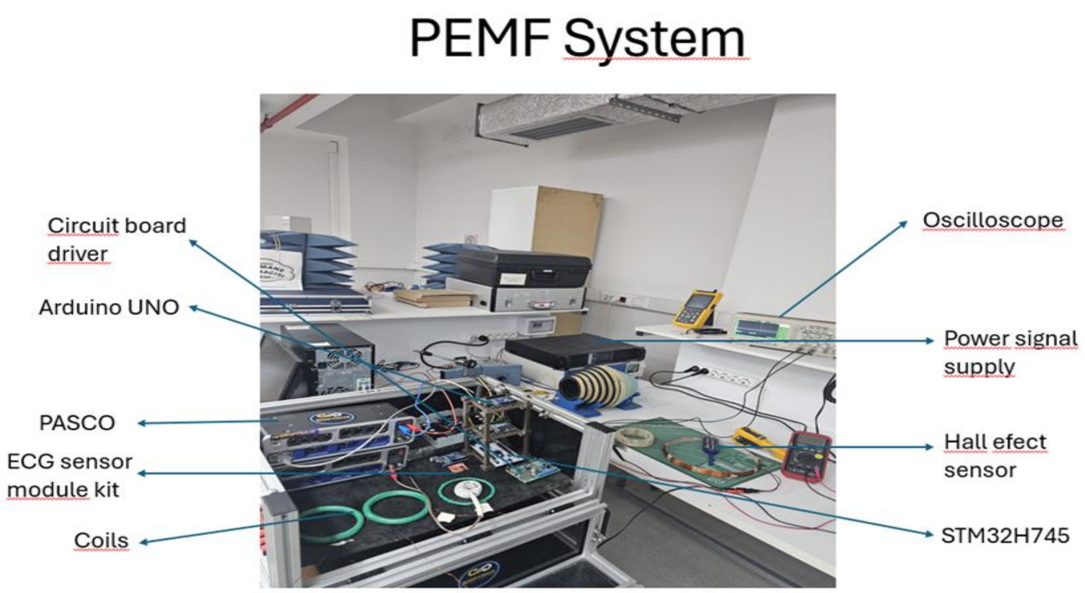
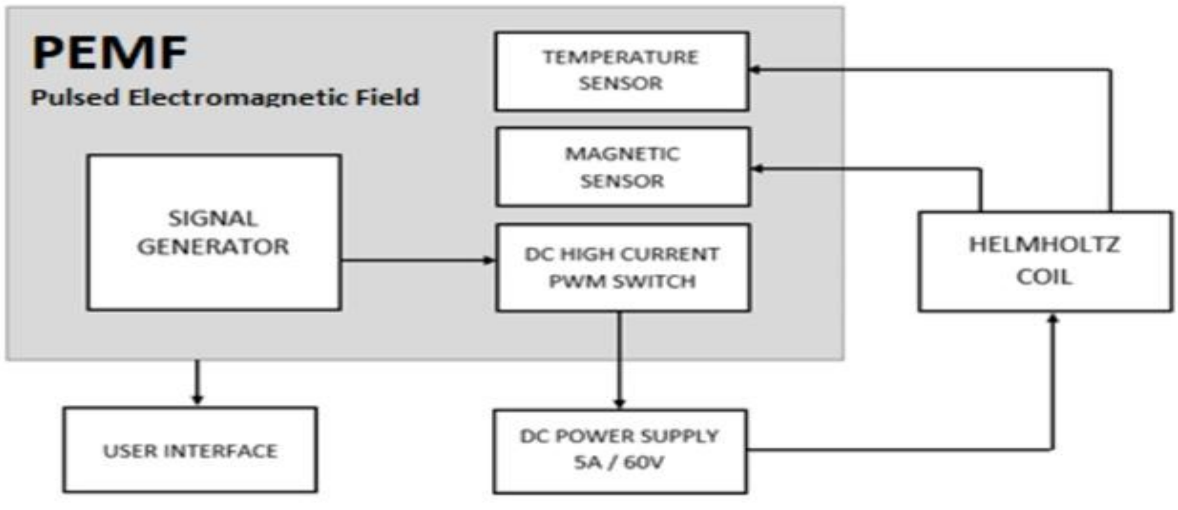


Figure 3 PEMF System



Principle block diagram of microcontroller-based pulsed electromagnetic fields (PEMF) device

Figure 4 Block diagram for PEMF

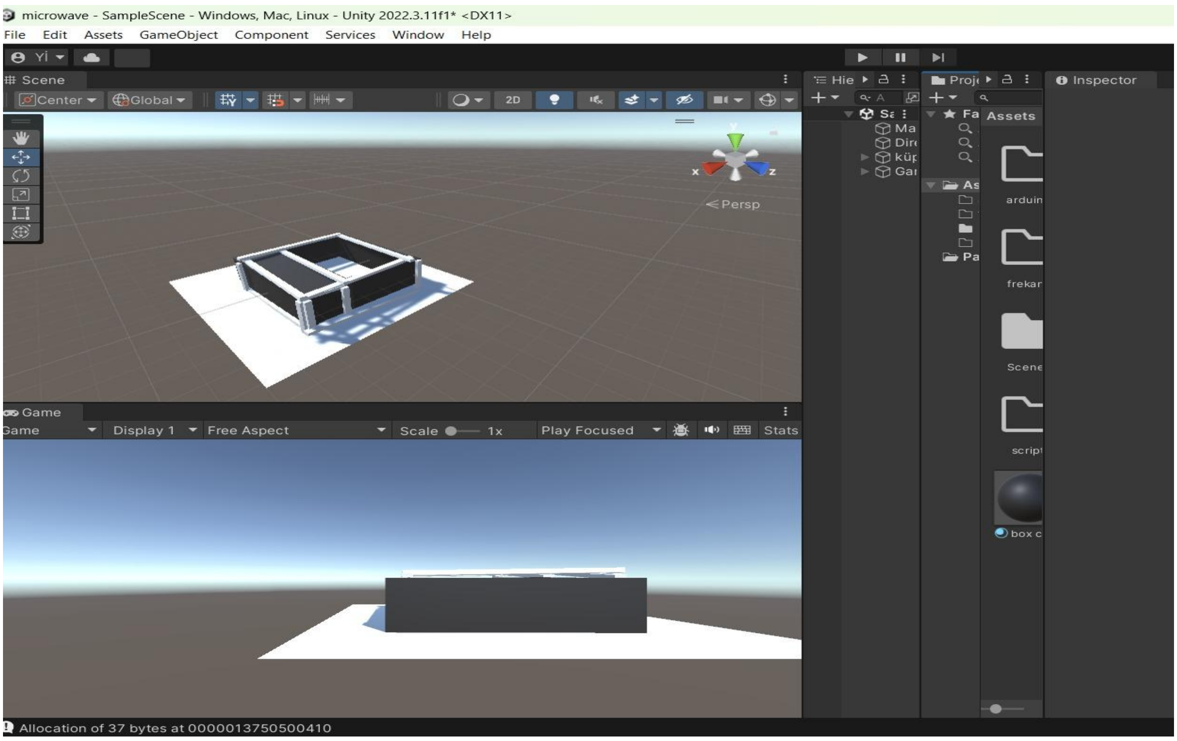


Figure 5 : First Phase of Digital Twin Design of PEMF Therapy System

This figure represents the first stage of the digital twin modeling of the PEMF (Pulsed Electromagnetic Field) application system used in the treatment of kidney cancer in a laboratory environment, created in the Unity program. This design was created to visualize the concept developed by taking as reference the process of digital modeling of the structural and functional properties of the real system.

3.4. Testing Plan

Requirement	Testing
Selectable Waveforms of sine,square,triangle and sawtooth waveforms	Use oscilloscope to display if signal generator is outputting sine,square or triangle waveforms
Frequency range of 10 Hz to 1 Khz	Use oscilloscope to measure the frequency of the waveforms
Amplitude range of -5 to 5 V	Use oscilloscope to check the amplitüde of the waveform
30 Watt Amplifier	Attach equivalent impedance of the coil to amplifier,and measure voltage across to calculate power
Power coil with diameter of 4 cm and 15 cm	Use caliper to measure diameter of coils
Digital voltage display with range of 0 to 300 AC V	Attach function generator,and vary frequencies and ampolitudes.Replace built digital voltage display with actual DVM to test accuracy,and compare numbers between out built digital voltage display and actual DVM
Magnetic field below 1 T	Magnetic field from 1-15 mT
Detector coil to measure magnetic flux	Use gauss meter to test accuracy of the detector coil
Dc voltage source of 30 V	Use digital volt meter tı measure DC voltage
Weight less than 15 lbs	Use scale

Table 1:Testing Plan

IV. SIMULATION AND RESULTS

4.1 CST Simulation With Petri Dish

The coil properties used in CST simulation are: current = 1.5 A, number of windings = 200, resistance = 9 Ω, radius = 120 mm, thickness = 15 mm.

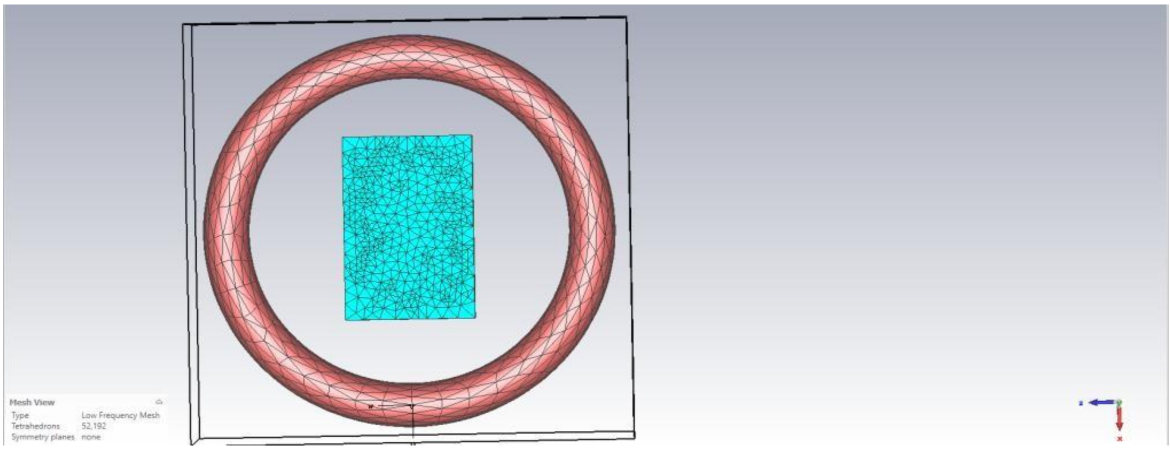


Figure 6: Mesh View

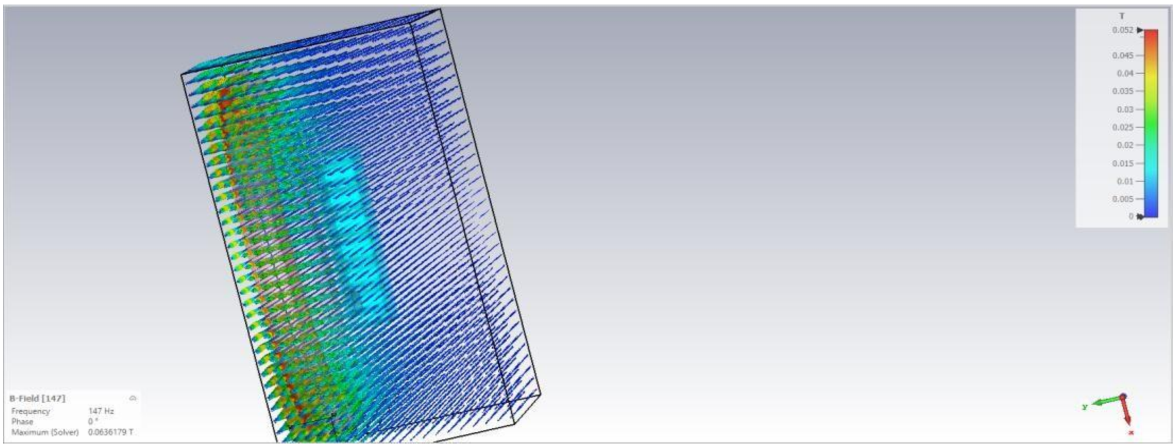


Figure 6.1:B-Field

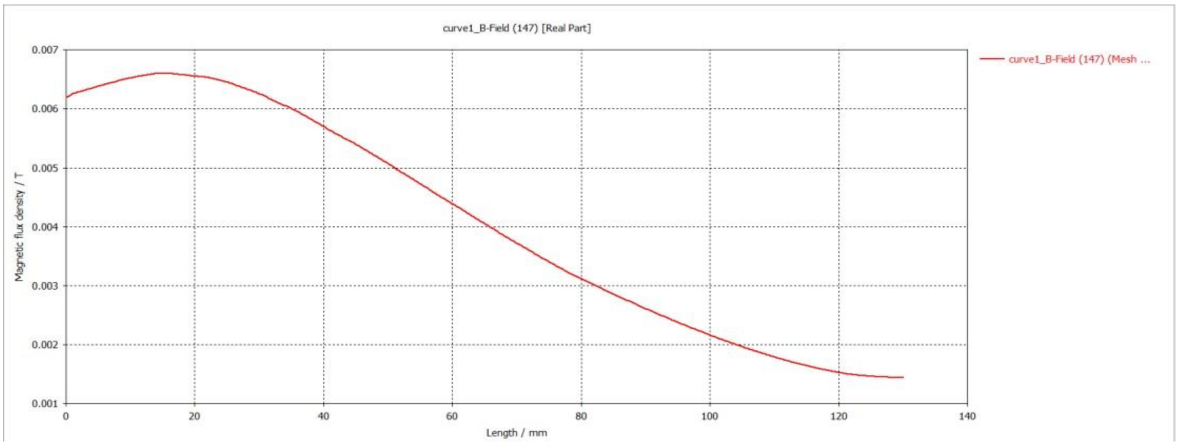


Figure 6.2:B-Field on the line

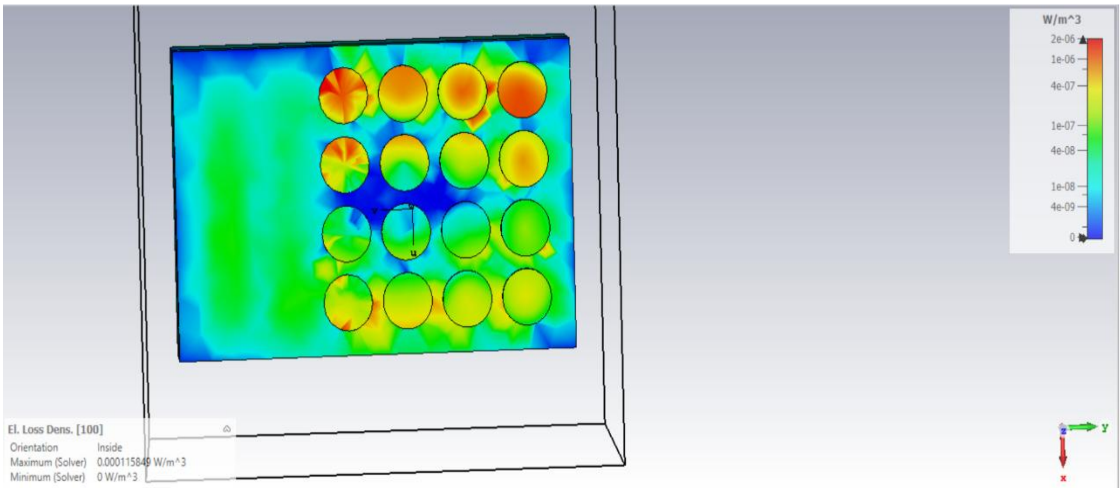


Figure 6.3: Electric loss density

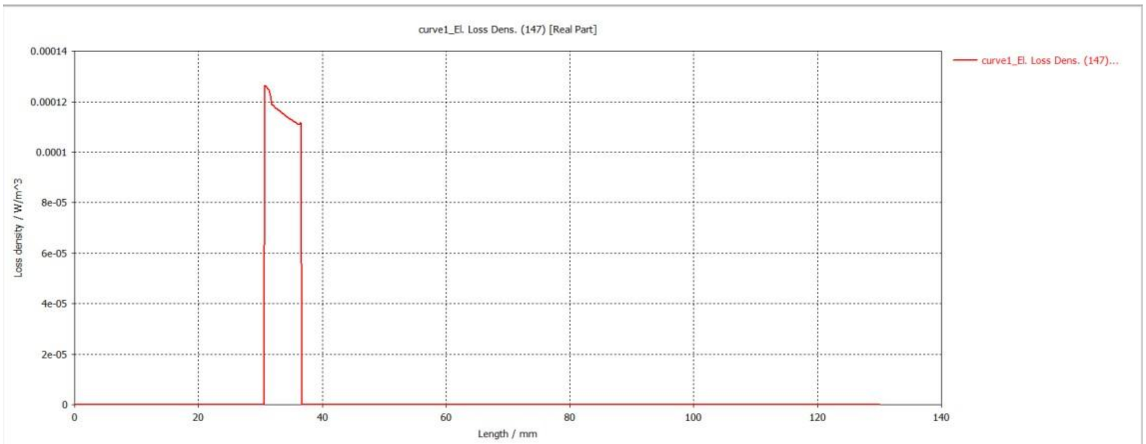


Figure 6.4: Electric loss density on the line

4.2 Result of CST Simulation with Petri Dish

In the CST simulation, a coil with an empty assembly was placed, and a petri dish was placed 3 cm in front of it. The first eight circles of the petri dish contained tumor cells. The remaining eight contained healthy cells. A line was drawn from the coil through any tumor in the petri dish. The B-field (magnetic field) and electrical loss intensity were observed. Based on the observations, the B-field along the drawn line began at 6 mT (millitesla) and decreased toward 0 with distance from the coil. A clear increase in electrical loss intensity was observed as soon as the line entered the petri dish. This was due to the tumor's set values of 30,000 epsilon and 0.05 electrical permittivity.For healthy cells, the values of 10,000 epsilon and 0.001 permeability were entered.

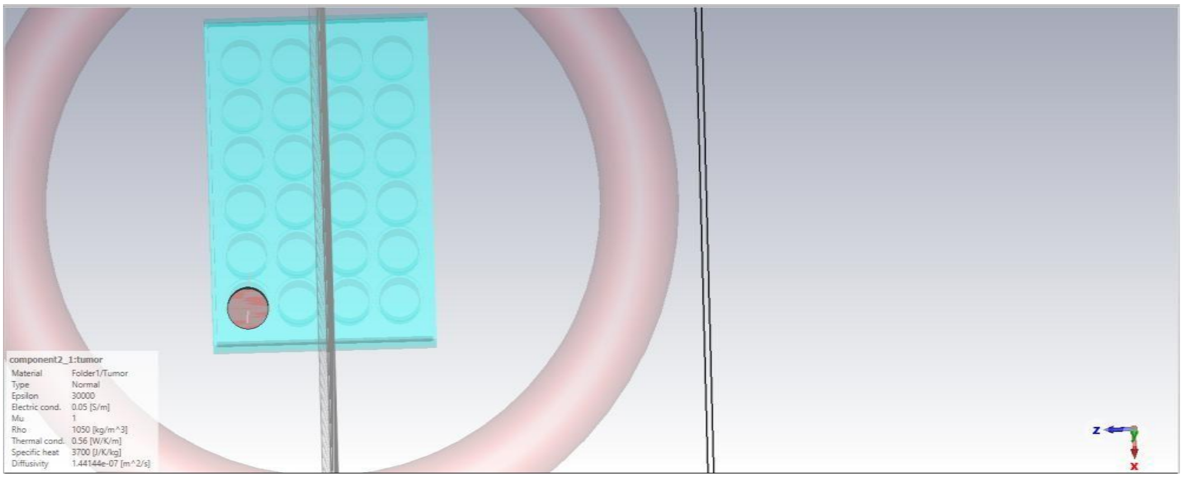


Figure 7: Material properties for tumor

4.3 CST Simulation With Kidney(3D Slicer)

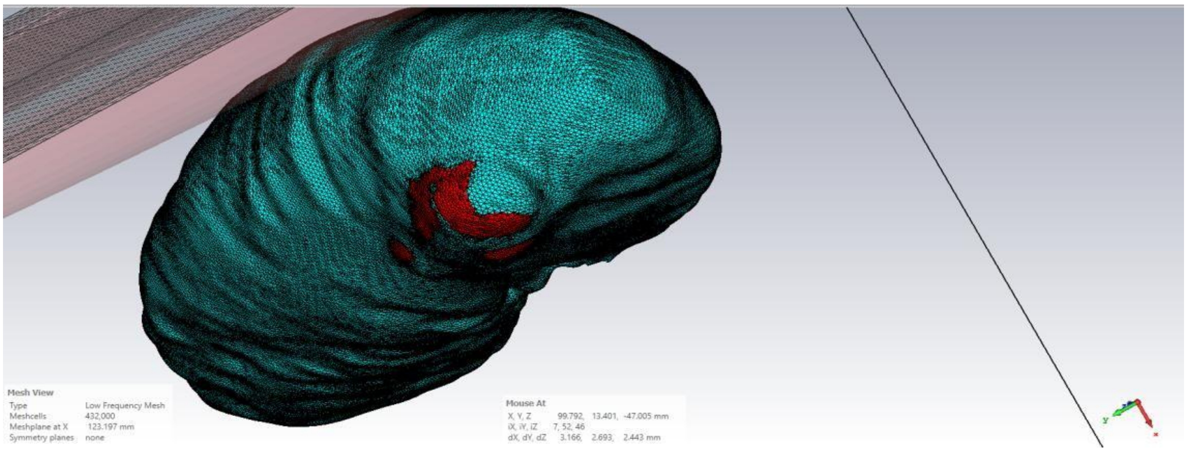


Figure 7.1: Kidney Mesh view

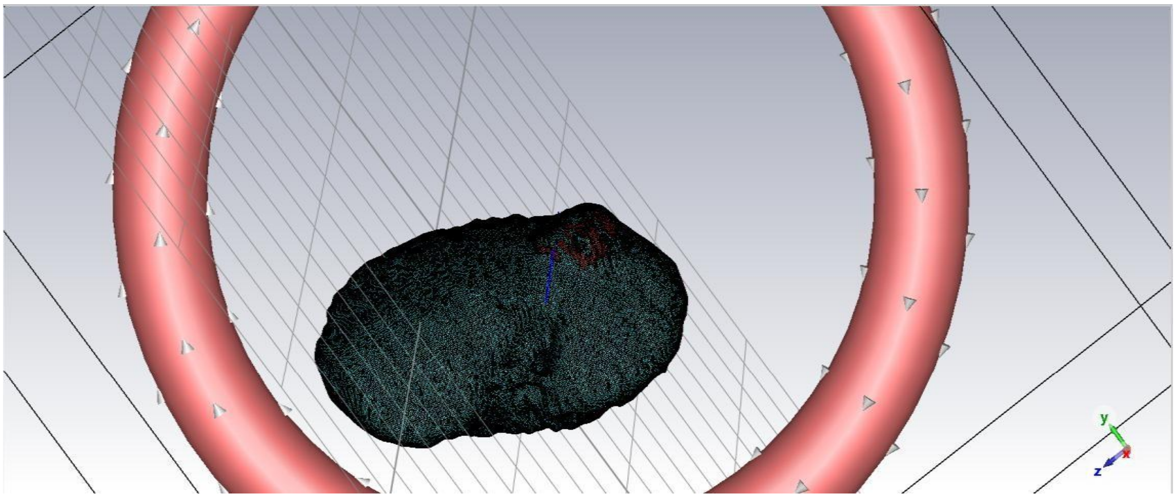


Figure 7.2: CST Simulation With Kidney(147hz)

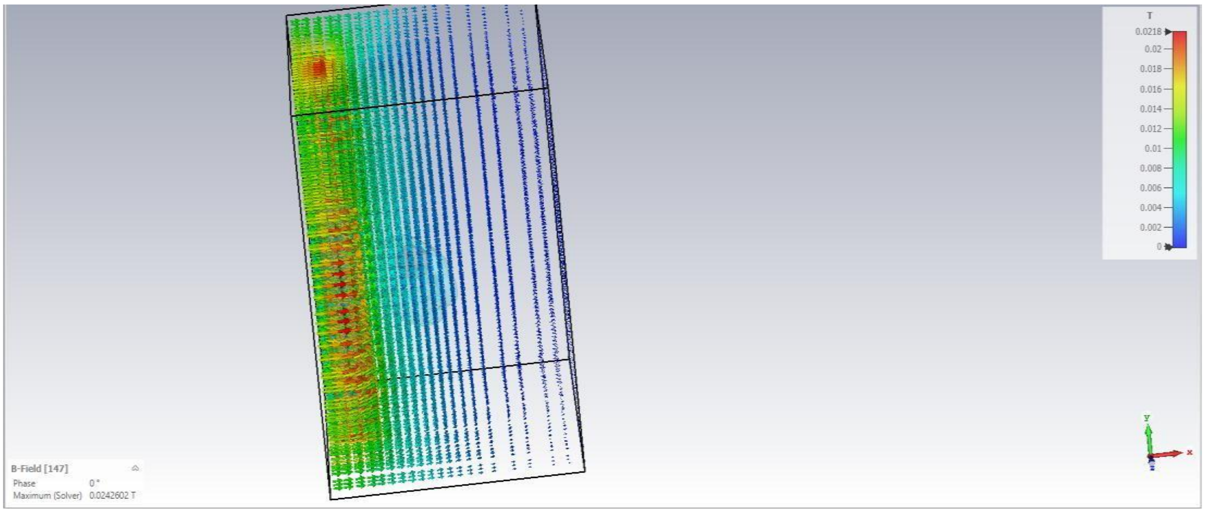


Figure 7.3: Kidney B Field

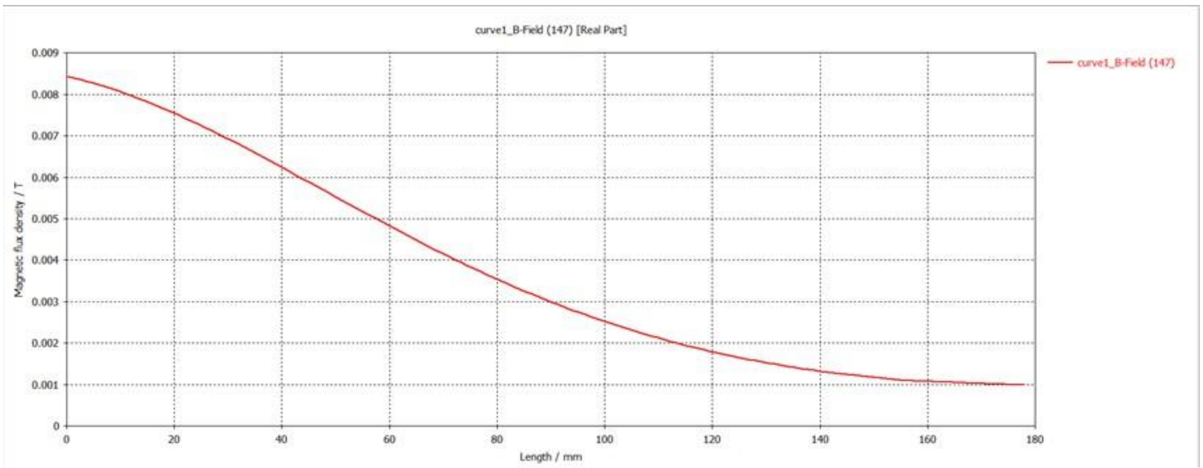


Figure 7.4: Kidney B Field on the line

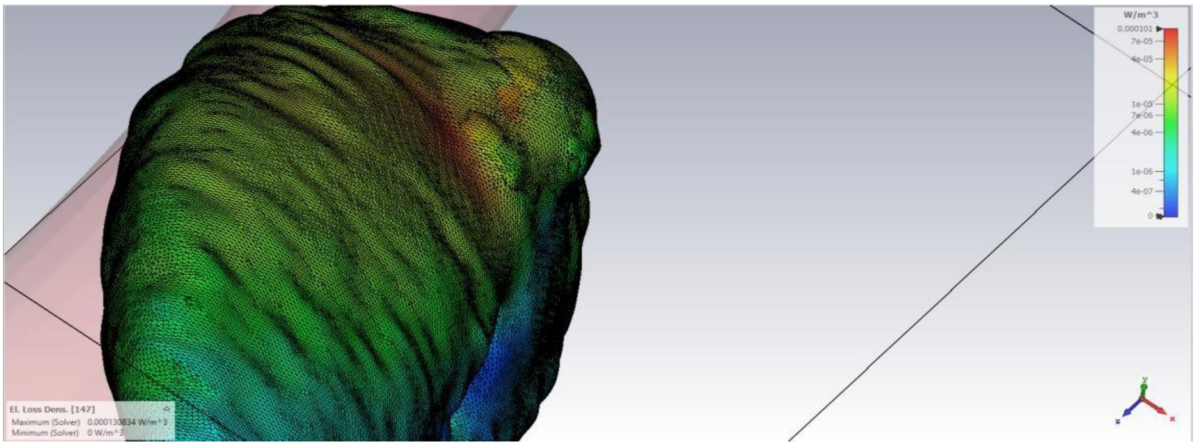


Figure 7.5: Kidney Electric loss density

4.4 Results of CST Simulation with Kidney

In the CST simulation, a coil with an empty assembly was placed, and a kidney with a tumor, taken from a 3D slicer, was positioned 3 cm in front of the coil. A line was drawn from the coil through the tumor in the kidney. The B-field (magnetic field) and electrical loss intensity were observed. Based on the observations, the B-field along the drawn line began at 8 mT (millitesla) and decreased towards 0 with distance from the coil. The electrical loss intensity of the kidney was observed to be significantly more intense in the tumor region. A significant increase in electrical loss intensity was observed when the line entered the tumor. This is due to the dielectric constant of 30,000 and permittivity of 0.05 determined for the tumor, compared to the dielectric constant of 10,000 and permittivity of 0.01 determined for the healthy kidney.

Tissue Type	Dielectric Constant (ε')	Loss Factor (tan δ)	Conductivity (σ, S/m)	Frequency
Healthy Tissue	10,000 - 15,000	0.1 - 0.2	0.001 - 0.01	100 Hz
Cancerous Tissue	20,000 - 30,000	0.2 - 0.5	0.01 - 0.05	100 Hz

Figure 8: Material Properties of Healthy and Cancerous Tissues

4.5 Phantom Model Experiment

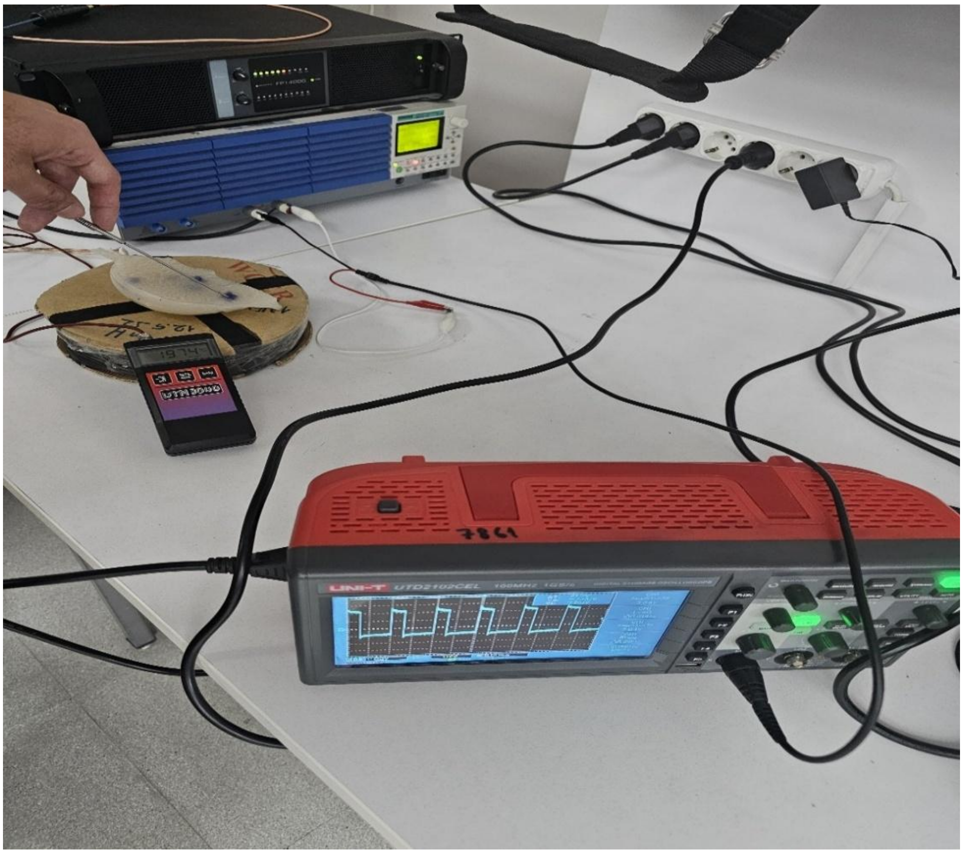


Figure 9: Phantom Model Experiment

Signal Type	Time(Minutes)	Frequency(Hz)	Temperature inn healthy tissue(°)	Temperature in cancerous tissue(°)	Magnetic Field in iron powder spilled cancerous tissue	Magnetic Field in iron powder unshed cancerous tissue	Magnetic field in healthy tissue	Current(A)	Voltage(V)	Power (W)

Puls,35	0	100	11	12	5 mT	4.5 mT	4.2 mT	2	4	5.6
Puls,35	10	100	11.4	14.67	5 mT	4.5 mT	4.2 mT	2	4	5.6
Puls,35	20	100	11.9	17	5 mT	4.5 mT	4.2 mT	2	4	5.6
Puls,35	30	100	12.7	19.6	5 mT	4.5 mT	4.2 mT	2	4	5.6

Table 2: Results of the Phantom Model Experiment

Temperature Increase and Magnetic Field Analysis
• Temperature increases were measured at different magnetic field intensities and frequencies.
• In the experiment, it was observed that the temperature increase was more pronounced in cancerous tissue.
• It was observed that the magnetic field had relatively lower effects in healthy tissue and cancerous tissues without iron dust.

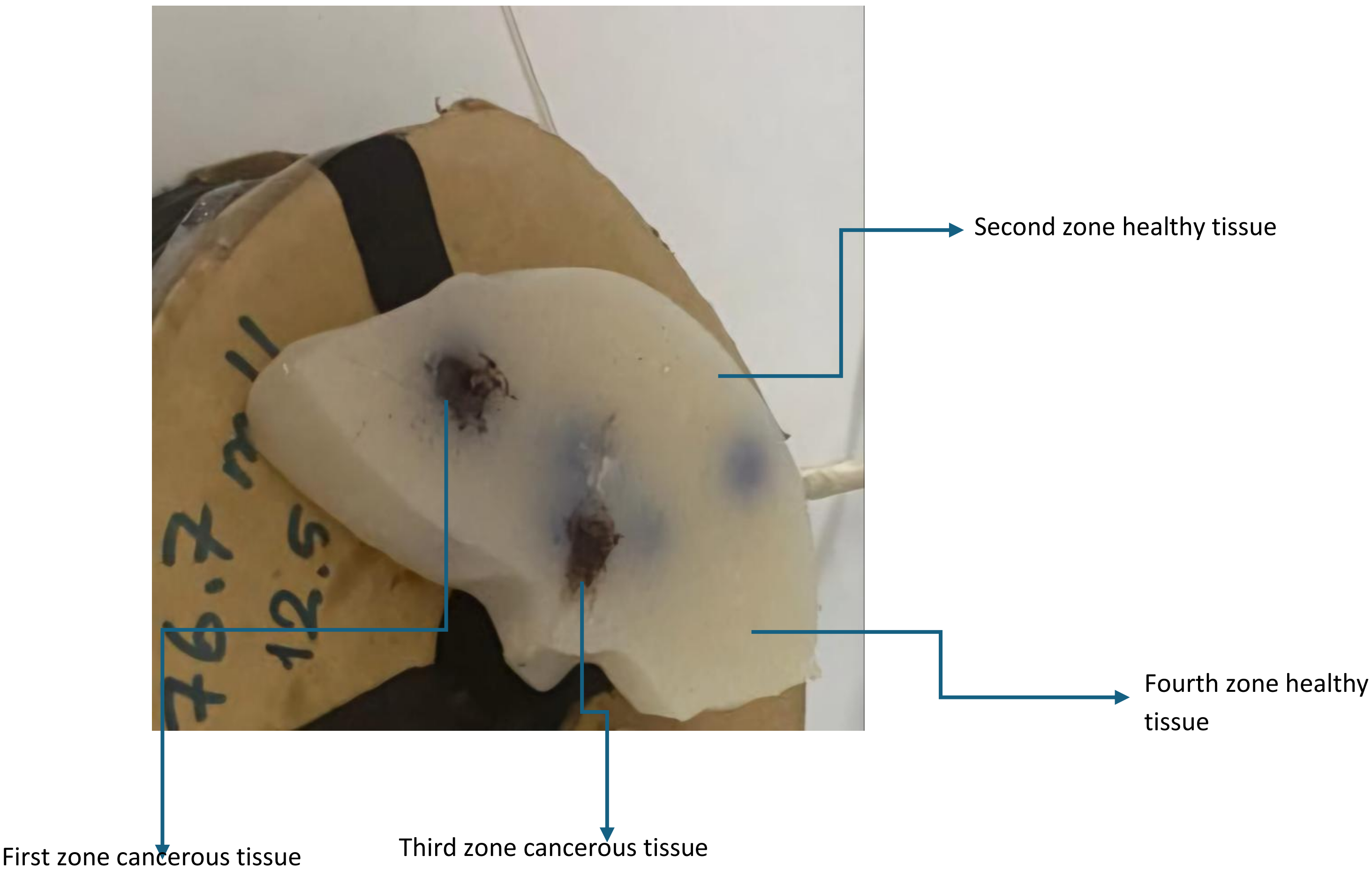


Figure 10: Phantom model of the kidney

Signal type	Time(Minutes)	Frequency(hz)	The temperature in the first zone cancerous tissue(°)	The temperature in the Second zone healthy tissue(°)	The temperature in the Third zone cancerous tissue(°)	The temperature in the Fourth zone healthy tissue(°)
Puls,35	0	100	18	19	18	20.2
Puls,35	10	100	20.1	21	19.6	20.5
Puls,35	20	100	21.9	21.4	21	20.7
Puls,35	35	100	23.5	22	22.4	21

Table 3: Temperature change table of different regions in the kidney phantom model

Temperature change analysis

- The temperature increases observed in this experiment indicate that cancerous tissues can be heated more by the application of magnetic fields compared to healthy tissues and can be used for targeted therapy.

4.6 Kidney Cancer Cell Application Results

	0 h	12 h	24 h	48 h	72 h
CONTROL					

75 Hz					
100 Hz					
150 Hz					

Table 4 : Change in kidney cancer cells with respect to time in different frequency applications and comparison with control group cells

Dark areas (Focal Points) represent the density or structural integrity of cancer cell populations. Blurring/diffuseness indicates signs of post-treatment cell integrity, cell destruction, or apoptosis. Compared to the control group, 150-hertz frequency applications yielded better results.

V. CONTROL OF THE SYSTEM WİTH GUI AND EMBEDDED SYSTEM DESIGN

5.1 Gui for PEMF system

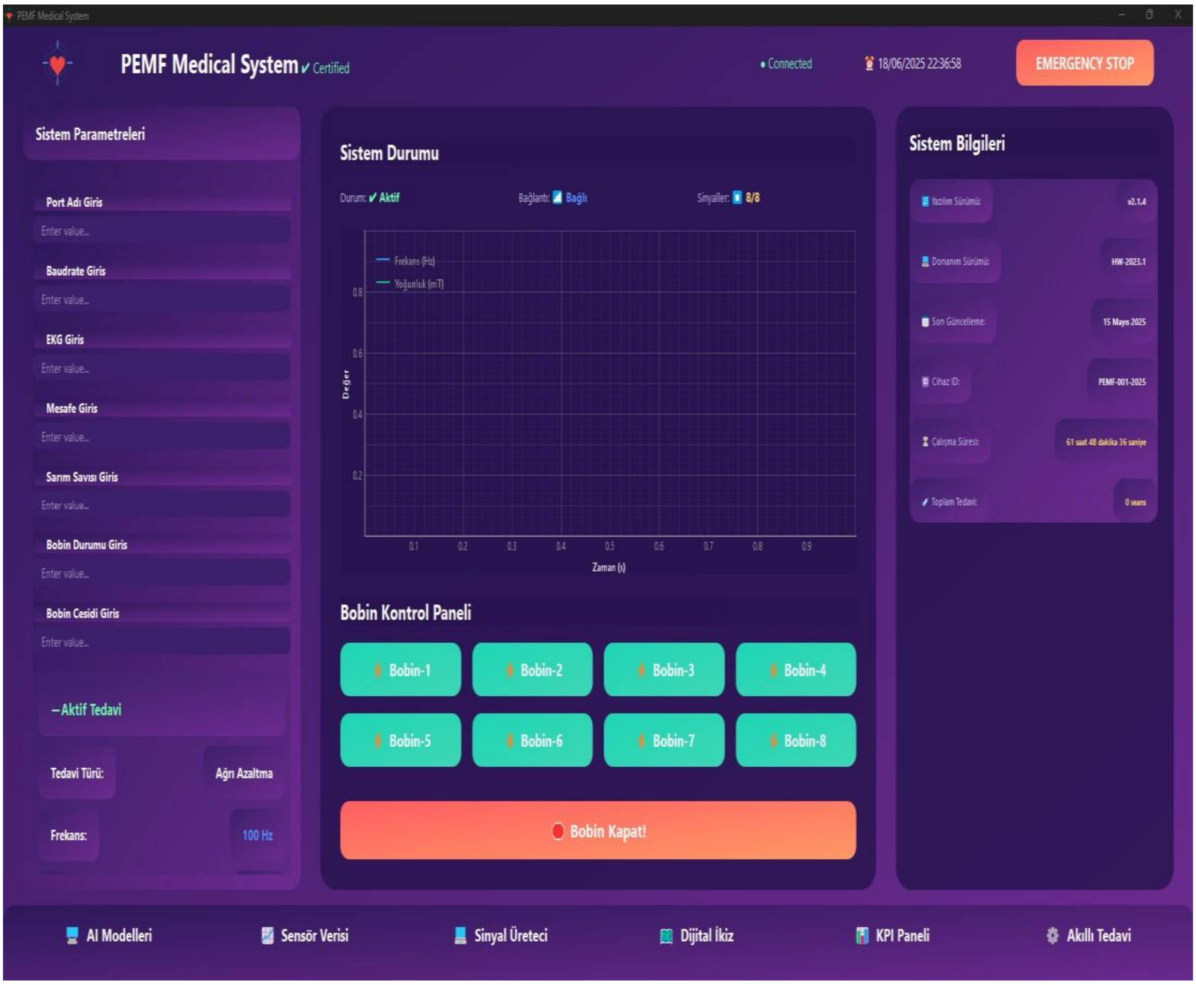


Figure 11: Gui main screen

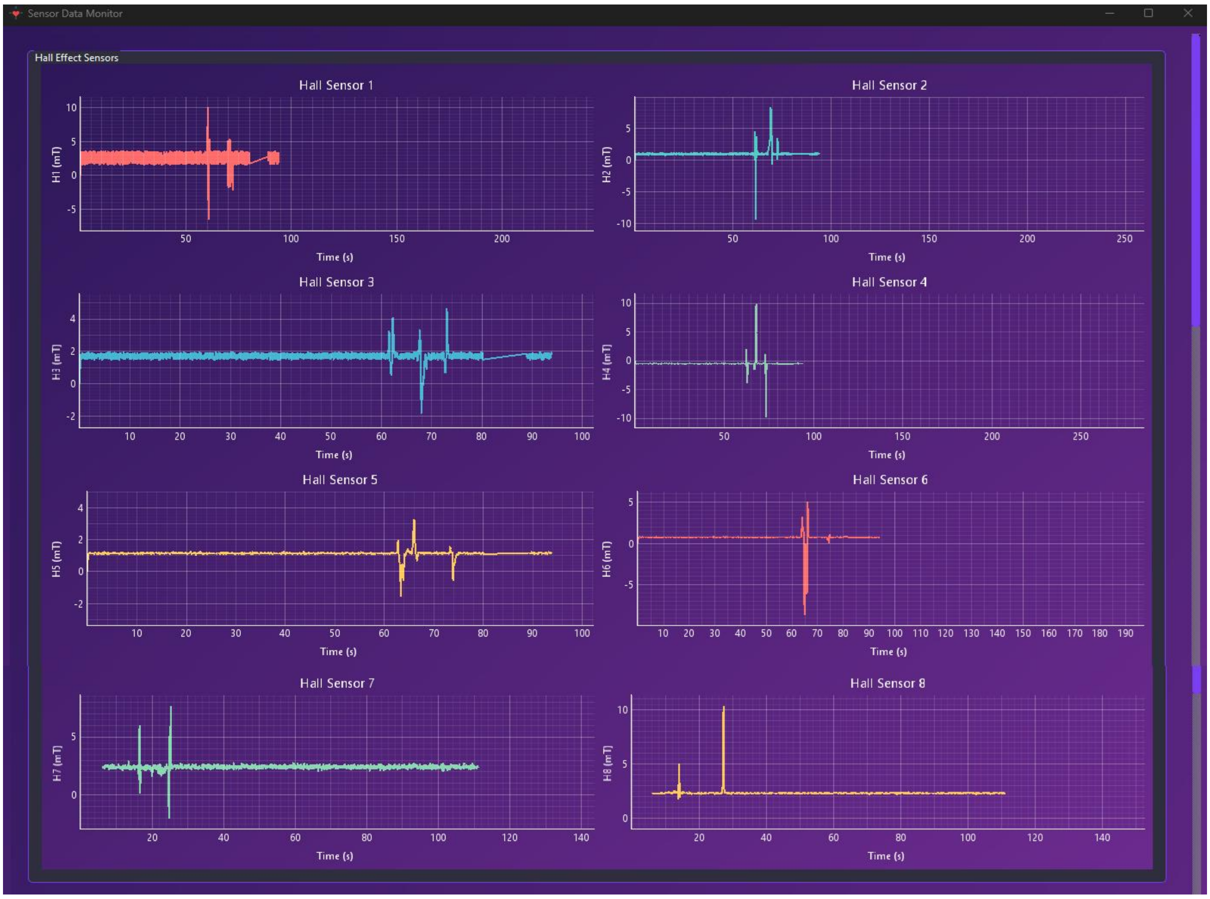


Figure 12: Sensor data window(Hall effect sensors)

Figure 13: Sensor data window(Current sensors and Temperature&Humidity sensor)

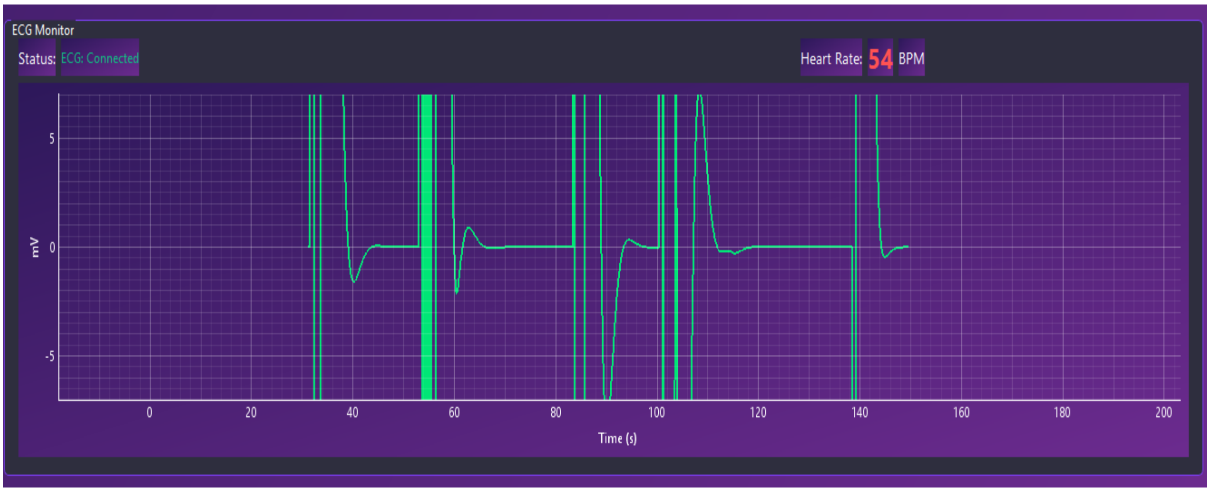


Figure 14: Sensor data window(Heart rate Sensor)

This interface is designed to control the PEMF system. It has a coil control panel, AI support, sensor data graphs, PWM control, and many other functions. In Figure 12, a separate magnet was placed near each magnetic field sensor to test it.

Figure 15: PWM Signal Generator

Eight separate timers of the STM32 card can be controlled from the PWM control generator section. PWM signals can be generated by adjusting the frequency and duty cycle values of each channel.

5.2 Embedded system design

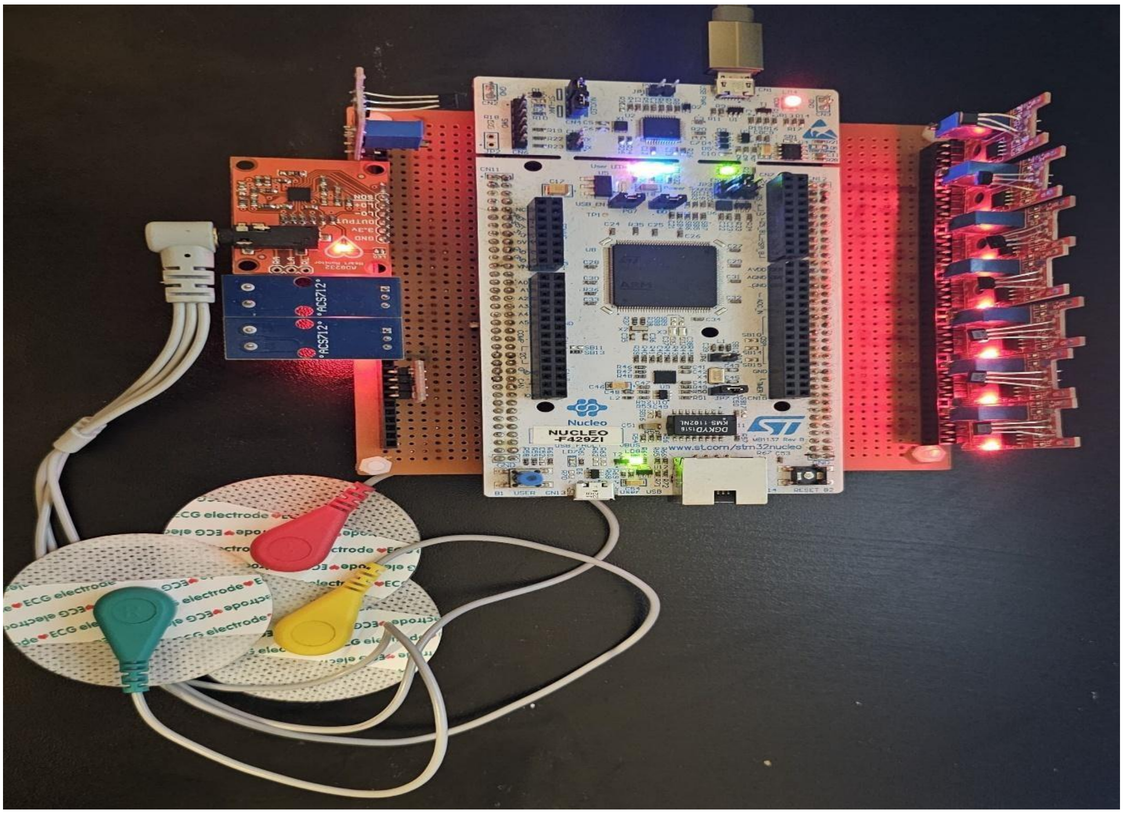


Figure 16: Microcontroller for PEMF System

5.2.1 Working logic of sensors

Hall effect sensors:
```
float voltage1, magneticField1;
float calculateMagneticField1(uint32_t adcValue) {
    voltage1 = ((float)adcValue / ADC_RESOLUTION) * HALL_VDD;
    magneticField1 = (voltage1 - HALL_ZERO_VOLTAGE) / HALL_SENSITIVITY[0];
    return magneticField1;
}
```
This function converts the digital value from the ADC first into voltage and then into magnetic field strength using the Hall sensor's calibration information. Thus, the sensor output is obtained directly in the physical unit mT.

Current sensors:
```
float calculateCurrent(uint32_t adcValue, float zeroPoint) {
    // If ADC reading is too far from expected value, use the expected value
    if(fabsf(adcValue - ACS712_ADC_OFFSET) > 100) {
        adcValue = ACS712_ADC_OFFSET;
    }

    // Convert ADC value to voltage using current sensor supply voltage
    float voltage = (adcValue / ADC_RESOLUTION) * CURRENT_VDD;

    // Calculate current based on ACS712 sensitivity and calibrated zero point
    float rawCurrent = (voltage - zeroPoint) / ACS712_SENSITIVITY;
```

```
    // Apply noise filtering and limit the current range
    if(fabsf(rawCurrent) < ACS712_NOISE_THRESHOLD) {
        return 0.0f;
    }
```

This function first converts the ADC value from the ACS712 current sensor to voltage and then calculates the current using the zero point as a reference. A noise filter and drift control prevent erroneous measurements and return a more accurate current value.

Temperature&Huminity sensor:

```
HAL_StatusTypeDef SHT31_Init(void)
{
  uint8_t cmd[2] = {0x30, 0xA2};
  HAL_I2C_Master_Transmit(&hi2c1, SHT31_ADDR, cmd, 2, 100);
  HAL_Delay(10);
  return HAL_OK;
}

HAL_StatusTypeDef SHT31_ReadTemperatureHumidity(float *temperature, float
*humidity)
{
  uint8_t cmd[2] = {(SHT31_MEAS_HIGH >> 8) & 0xFF, SHT31_MEAS_HIGH & 0xFF};
  HAL_I2C_Master_Transmit(&hi2c1, SHT31_ADDR, cmd, 2, 100);
  HAL_Delay(20);
  HAL_I2C_Master_Receive(&hi2c1, SHT31_ADDR, sht31_rx_data, 6, 100);

  if (!SHT31_CheckCRC(&sht31_rx_data[0], 2, sht31_rx_data[2]) ||
      !SHT31_CheckCRC(&sht31_rx_data[3], 2, sht31_rx_data[5]))
    return HAL_ERROR;

  uint16_t temp_raw = ((uint16_t)sht31_rx_data[0] << 8) | sht31_rx_data[1];
  uint16_t hum_raw = ((uint16_t)sht31_rx_data[3] << 8) | sht31_rx_data[4];

  *temperature = -45.0f + 175.0f * temp_raw / 65535.0f;
  *humidity = 100.0f * hum_raw / 65535.0f;

  return HAL_OK;
}
```

This function initializes the SHT31 temperature-humidity sensor and reads temperature and humidity data by sending a measurement command. The raw data is converted into physical values according to the sensor formula and returned as output.

Heart rate sensor:

```
void readAD8232(void) {
    // LO+ ve LO- pinlerini kontrol et (PF6 ve PF5)
    uint8_t lo_plus = HAL_GPIO_ReadPin(GPIOF, GPIO_PIN_6);
    uint8_t lo_minus = HAL_GPIO_ReadPin(GPIOF, GPIO_PIN_5);

    if (lo_plus || lo_minus) {
        // Elektrot yerinden çıkmış
        sprintf(uartBuffer, "Leads off detected!\r\n");
```

```
        HAL_UART_Transmit(&huart3, (uint8_t*)uartBuffer, strlen(uartBuffer),
UART_TIMEOUT);
        return; // ADC okumayı yapma
    }

    ADC_ChannelConfTypeDef sConfig = {0};
    sConfig.Channel = ADC_CHANNEL_14;  // PF4
    sConfig.Rank = 1;
    sConfig.SamplingTime = ADC_SAMPLETIME_144CYCLES;

    if (HAL_ADC_ConfigChannel(&hadc3, &sConfig) == HAL_OK) {
        HAL_ADC_Start(&hadc3);
        if (HAL_ADC_PollForConversion(&hadc3, 100) == HAL_OK) {
            ad8232_adc_value = HAL_ADC_GetValue(&hadc3);
            ad8232_voltage = (ad8232_adc_value / ADC_RESOLUTION) * AD8232_VDD;

            // Store in circular buffer
            ad8232_buffer[ad8232_buffer_index] = ad8232_adc_value;
            ad8232_buffer_index = (ad8232_buffer_index + 1) % AD8232_BUFFER_SIZE;
        }
    }
}
```

This function checks whether the electrodes are connected to the AD8232 ECG sensor. If so, it reads the heart signal data via the ADC and converts it to voltage. The reading is stored in a circular buffer.

5.2.2 Magnetic Field Generation and Measurement with PWM Controlled Coil System

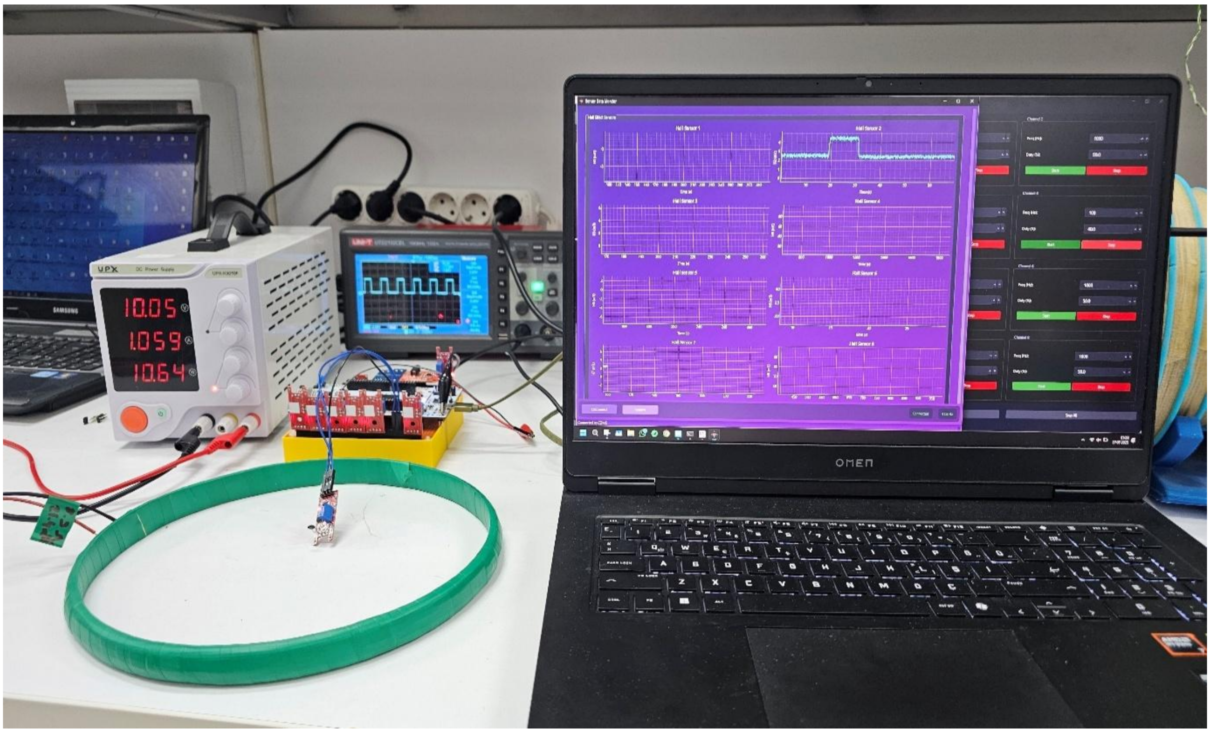


Figure 17:PEMF System

The 4th PWM channel in STM32 was set to 100Hz and 40% duty cycle. Coil properties: number of windings = 200, resistance = 8.9Ω, radius = 120mm, thickness = 15mm. The coil powered by 10V worked with 1.059A current and consumed 10.64W power. The magnetic field values were measured at the center and corner of the coil with one of the magnetic field sensors. The magnetic field was measured as 2.5mT at the center of the coil and 4mT at the corner.

VI. DISCUSSION IN TERMS OF CONTRAINTS

6.1 Ethical Constraints

In our project, the integration of AI-based tuning mechanisms is designed to ensure that system operations do not interfere with regulated frequency allocations or introduce any form of unfair bias in system behavior. Although the AI functions at the analog control level, responsible design practices have been implemented through transparent decision-making algorithms. Furthermore, the system avoids the use of any personal or user-identifiable data, thereby upholding stringent privacy standards. The development process is fully aligned with IEEE's ethical design principles and emphasizes the protection of user privacy, particularly when the device is deployed in clinical environments involving direct human interaction.

6.2 Health Constraints

The coil systems employed in this study are optimized to generate electromagnetic fields (EMFs) that are effective for kidney cancer treatment while operating within safe power limits. Although the output power is configured to achieve the desired therapeutic effect, a comprehensive assessment of exposure levels has been conducted to ensure that all generated EMF intensities remain well within ICNIRP safety guidelines. This rigorous analysis confirms that the system poses no health hazard to patients, clinical operators, or individuals in the surrounding environment, thereby safeguarding overall human health during industrial or clinical deployment.

6.3 Legal Constraints

Our system is developed in full compliance with both regional and international regulations governing medical devices and electromagnetic field applications. Adherence to standards such as ISO 10974, IEC 60601-1, and other applicable FDA/CE requirements has been a fundamental aspect of the design process. The system is engineered to maintain transmitted power, operational frequency ranges, and modulation characteristics within legal limits, ensuring that every phase of development and subsequent clinical deployment meets the rigorous criteria set by regulatory bodies.

6.4 Security Constraints

Although this project primarily focuses on the hardware-level control and optimization of coil designs, significant measures have been taken to mitigate security risks that may arise from wireless data transmission and control interfaces. The design incorporates robust EMI shielding, careful PCB layout, and thorough ground plane isolation to minimize vulnerabilities to external interference or potential malicious tampering. In anticipation of future enhancements, provisions for integrating secure communication protocols—such as encrypted UART or SPI—are being considered to further protect data integrity and reinforce overall system security.

These constraints collectively ensure that our system not only meets the technical and clinical requirements for effective cancer treatment but also adheres to the highest ethical, health, legal, and security standards.

VII. STANDARTS

The AI-assisted coil design system for kidney cancer treatment is developed in strict accordance with several international and industry-specific standards to ensure electromagnetic compatibility (EMC), patient safety, reliable performance, and seamless clinical integration. Compliance with these standards was carefully considered during the design, simulation, testing, and validation phases to enable legal deployment and industrial viability.

- **ISO 10974:** This standard provides comprehensive guidelines for the safety of electromagnetic fields in medical environments. In our project, ISO 10974 governs the design criteria for the EMF exposure levels generated by the coil systems. The coils are optimized to ensure that the therapeutic electromagnetic field intensities remain within the specified safe limits for patient application, thereby mitigating any risk of adverse tissue effects.
- **IEC 60601-1:** As the fundamental standard for medical electrical equipment, IEC 60601-1 outlines the essential safety and performance requirements for clinical devices. Throughout the development process, our system has been engineered to adhere to these strict guidelines. Detailed testing ensures that all electrical and mechanical components, as well as the integrated AI control mechanisms, meet the rigorous standards for patient safety and overall device reliability.
- **IEC 60601-1-2 (EMC Requirements):** To ensure robust electromagnetic compatibility, our PCB design and system integration incorporate strategies such as EMI shielding, controlled grounding, and the use of decoupling capacitors. These measures, in line with IEC 60601-1-2, guarantee that the system can operate reliably in complex clinical environments without causing or suffering from electromagnetic interference.
- **IEEE 11073:** The IEEE 11073 standard facilitates effective medical device communication, ensuring interoperability and efficient data exchange between the AI system and clinical monitoring equipment. By following this standard, our design guarantees that real-time patient data and system diagnostics can be seamlessly integrated into broader clinical information systems, enhancing overall treatment management.
- **ISO 13485:** Although currently in the research and development phase, our design process aligns with the principles of ISO 13485, which governs quality management systems for medical devices. This structured approach ensures traceability, continuous improvement, and technical reliability—key factors that support future industrial production and certification efforts.
- **IEC 62304:** For the software aspects of our project, particularly the Python-based AI algorithms used for system tuning and real-time control, adherence to IEC 62304 is maintained. This standard outlines the software lifecycle processes that ensure the safety and effectiveness of medical device software, thereby supporting robust development practices and regulatory compliance.

Through meticulous adherence to these standards, the developed coil design system not only meets stringent academic and technical expectations but also establishes a solid foundation for integration into certified clinical infrastructures. The comprehensive standard compliance ensures that the system operates safely, effectively, and in full alignment with international regulatory requirements, thereby paving the way for its successful industrial deployment in the healthcare sector.

VIII. SUMMARY AND CONCLUSİON

In this study, a customizable Pulsed Electromagnetic Field (PEMF) device—referred to as the DMAT system—was developed specifically for kidney cancer treatment. The system integrates a range of coil geometries optimized via AI-supported algorithms to maximize magnetic flux density and field uniformity while operating within safe exposure limits (ICNIRP, ISO 10974). Selectable waveforms (sinusoidal, square, sawtooth) and frequencies (0–150 Hz) were implemented to identify the optimal parameters for inducing apoptosis in renal carcinoma cells. Numerical simulations in CST Studio Suite and laboratory measurements quantified the effects of coil winding number, wire thickness, and coil diameter on magnetic field distribution and energy consumption, while phantom model experiments and cell culture assays validated therapeutic performance. Real-time monitoring of treatment surface area, temperature, and field intensity was achieved through Hall-effect and temperature sensors, with data visualization and control provided via a Python-based AI-supported GUI. An STM32F429ZI microcontroller executed closed-loop control, generating PWM signals to adjust coil current, voltage, and frequency dynamically.

Experimental results demonstrated that temperature increases were more pronounced in cancerous tissue compared to healthy tissue, confirming the potential for targeted thermal effects. In the kidney cancer cell application, 150 Hz frequency yielded superior reduction in cell density and integrity relative to the control group. The AI-driven optimization system effectively recommended appropriate waveforms, frequencies, and intensities based on sensor inputs, ensuring treatment parameters remained within ICNIRP safety guidelines. The flexibility of the device and its customization by taking into account different cancer types and patient characteristics provided a significant advantage, enhancing the potential for personalized therapy. Ethical, health, legal, and security constraints were rigorously addressed, and compliance with standards such as ISO 10974, IEC 60601, IEEE 11073, and GDPR/HIPAA was maintained throughout development.

In conclusion, by adopting a hybrid methodology that couples numerical simulations, AI-driven optimization, and experimental validation, this thesis establishes a robust framework for personalized PEMF therapies in oncology. The developed DMAT system aims to reduce side effects and increase the rate of complete remission in patients by offering an alternative approach to existing treatment strategies. The successful integration of coil design optimization, real-time monitoring, and AI-supported control paves the way for more effective, safe, and patient-specific cancer treatments.

IX.APPENDIX D – STANDARTS

• **ISO 10974 – Assessment of Electromagnetic Field Safety for Medical Devices**: This standard provides guidelines to ensure that the electromagnetic fields generated by therapeutic devices do not pose any risk to patient health. In our system, ISO 10974 has been applied to verify that the electromagnetic field intensities produced by the coil designs remain within safe limits during cancer treatment applications.

• **IEC 60601-1 – Medical Electrical Equipment – General Requirements for Basic Safety and Essential Performance**: IEC 60601-1 sets out the essential safety and performance requirements for medical electrical equipment. Our AI-assisted coil design system complies with these requirements, ensuring that all electrical and mechanical components, as well as integrated control systems, meet the highest safety standards for clinical use.

• **ICNIRP 2020 – Guidelines for Limiting Exposure to Electromagnetic Fields**: The system has been rigorously designed and tested to ensure that the intensity of the electromagnetic fields produced during treatment remains well below the safe exposure limits outlined in the ICNIRP 2020 guidelines. These internationally recognized guidelines provide critical thresholds for non-ionizing radiation, and adherence to them guarantees that both patients and clinical operators are not exposed to harmful levels of electromagnetic energy during system operation.

• **ISO 13485 – Quality Management Systems for Medical Devices**: ISO 13485 establishes a framework for quality management in the design, development, and manufacturing of medical devices. The quality processes in our project align with ISO 13485 principles, ensuring consistent traceability, rigorous testing, and continuous improvement throughout the system's development lifecycle.

• **HIPAA/GDPR – Data Protection and Privacy Regulations**: In the development of our system, robust measures have been implemented to ensure full compliance with HIPAA and GDPR data protection standards. The design emphasizes the anonymization of patient data, stringent access controls, and secure transmission protocols. By strictly adhering to these privacy regulations, the system safeguards sensitive medical information, maintaining confidentiality and preventing unauthorized access throughout the data collection, processing, and storage phases.

• **IEEE 11073 – Standards for Medical Device Communication**: IEEE 11073 facilitates interoperable communication between medical devices and healthcare information systems. By following these standards, our system ensures seamless data integration, enabling the real-time transmission of patient and system data to clinical monitoring platforms and other connected devices.

Through adherence to these standards, the developed system not only meets stringent clinical and technical requirements but is also well-positioned for future industrial deployment and regulatory approval.

Restrictions

Electromagnetic Power and Frequency Limits: According to ICNIRP and IEEE 10974 standards, the magnetic field intensity applied to human tissue should not exceed 1 mT. The system output operates in the range of 0–100 Hz, but has been tested at a maximum frequency of 50 Hz to remain within safety limits.

Material Durability: NiTi alloy wire can operate at a maximum temperature of 60 °C without deformation. The PCB material used for coils begins to soften above 80 °C in long-term heating tests.

Sensor and Hardware Limitations: The measurement range of Hall effect sensors is limited to ±10 mT; therefore, additional sensor calibration is required in configurations where the magnetic field is extremely high. IoT temperature sensors operate between −40 °C and +125 °C; in non-laboratory field conditions, accuracy loss at low temperatures can be up to 2 %.

Patient Safety and Clinical Limitations: The duration of application to the patient is limited to 30 minutes; excessive exposure may increase the risk of thermal damage. The device should be operated only with IEC 60601-1 compliant outlet and grounding conditions in a medically approved clinical environment.

Conditions

Experimental Environment: Laboratory Temperature: 20 °C ± 2 °C; Humidity: 40–60 % RH. Electromagnetic Noise: Other magnetic field sources (motors, high current cables) in the environment should be kept at least 1 m away.

Physical Placement and Positioning: The distance between the coil and the tissue/phantom is set as a fixed 30 mm. 1 cm² point sensor cells are used for temperature and magnetic field measurements in four regions (cancer/cancer iron powder, healthy/healthy iron-free) on the phantom model.

Sensor Calibration and Operating Conditions: Hall effect sensors are calibrated at the 0 mT reference point before each experiment. Temperature sensors are cross-validated with a K-type thermocouple; error margin is ± 0.3 °C.

Software and AI Operating Conditions: The Python-based AI module runs on Raspberry Pi 4 for real-time data processing. Model updates are included in the dataset by retraining offline after every 10 trials.

BUDGET PLAN

In this section, the budget plan and cost analysis of “DEVELOPMENT AND OPTIMIZATION OF COIL DESIGNS FOR CHILDREN’S CANCER WITH ARTIFICIAL INTELLIGENCE ASSISTED PULSED ELECTROMAGNETIC FIELD SYSTEMS” are presented. The cost analysis includes general categories such as equipment, consumables, office supplies, and phantom model experiment. The budget plan is shown in Table B-1.

Table B-1 *Budget Plan of the Project*

Category	Estimated Cost (TRY)
Equipment	4,000
Consumables	4,000
Office Supplies	500
Phantom Model	3,500
Total	12,000

Table 5: Budget Plan of the Project

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VITA (ÖZGEÇMİŞ)

Place the text of your Vita.